

Benign Exclusion and Consensus-Test Limits in Heterogeneous Federated Learning

Sebahattin Gökçen Özden and Kadir Sarıkaya

Abstract—Server-side defenses in federated learning decide which client updates to accept. Under heterogeneous data, honest updates can look anomalous, yet evaluations often report only attacked rounds, leaving the cost of rejecting honest clients unmeasured. We measure this cost for eight rules on four datasets, three heterogeneity levels (Dirichlet $\alpha = 0.5, 0.1, 0.01$), and five attack families (unbalanced coverage; 2,125 of 2,130 planned runs completed). The rules include Fed-MDBSCAN-G, our multi-density clustering filter anchored at a geometric-median acceptance region. With no attacker, Multi-Krum excludes 32.2% of honest submissions in all nine clean conditions, FLAME 42.4–48.9%, and Fed-MDBSCAN-G up to 42.6% at $\alpha = 0.01$; final accuracy falls up to 65.55 percentage points below plain averaging. Multi-Krum always keeps 61 of 90 updates, yet a random 61 recovered its accuracy to within 1.3 points of plain averaging on MNIST and Fashion-MNIST; FLAME kept its minimum majority of 46 in 719 of 990 clean rounds. Under distance-constrained attacks, Multi-Krum rejected only honest clients in five of six conditions. We derive a condition under which Fed-MDBSCAN-G’s cluster-validation stage cannot reject any group of initially accepted updates. It held on 35 of 36 attacked five-step checkpoints but on 22 of 72 canonical ones; under the constrained probes, copied attackers instead bypassed the test through extreme density. Removing every stage after the first changed mean final accuracy by under 0.1 percentage points. These findings explain failures of the evaluated implementations, not geometric defenses generally. We recommend reporting honest-client exclusion and accuracy loss without attackers alongside attack detection and success.

Index Terms—Federated learning, poisoning, robust aggregation, data heterogeneity, false positives, clustering defenses.

I. INTRODUCTION

FEDERATED learning trains a shared model from locally computed updates without collecting clients’ training examples at the server [1]. The same separation allows malicious clients to submit poisoned updates or to train on manipulated data [2], [3], [4]. Server-side defenses respond with robust aggregation, update filtering, or validation against a trusted reference [5], [6], [7], [8].

A filtering defense must answer two separate questions: does it exclude attackers, and does it preserve useful honest contributions? Statistical heterogeneity makes both harder. Honest clients can produce updates far from a round’s central

tendency, while coordinated attacks can stay close to the benign distribution [9], [10]. Prior evaluations have documented false positives, degraded utility, and backdoor vulnerability under these conditions [11], [12], [13]. These findings call for evaluating defenses without attackers and examining how each stage of a composite defense contributes to its final decision.

This cost matters in practice. In cross-device and Internet-of-Things deployments, the clients whose data differ most from the majority, such as those holding rare classes or unusual sensor profiles, are the ones most likely to submit updates that look anomalous. A defense that rejects them can trade coverage and fairness for robustness even when no attacker is present [11], [14], [15]. Our measurements count how many honest submissions are excluded in each round; they do not identify which clients are excluded repeatedly or measure the effect on individual clients. Evaluations that report detection and attack success only in attacked rounds leave this trade-off unmeasured [16]. Our own starting point was a design attempt: we adapted a multi-density clustering algorithm to federated filtering, expecting it to recognise heterogeneous honest groups rather than reject them. When its benign exclusion reached tens of percent under strong heterogeneity, the question became why, that is, which stage of the composite excludes honest clients and whether later stages can undo that decision.

We study these questions using a completed evaluation of eight server-side rules and a detailed analysis of Fed-MDBSCAN-G (Fig. 1), a composite filter that we designed and that combines a geometric-median acceptance region, relative-density partitioning, and shared-neighbour cluster validation. Our aim is to explain the costs and failures observed in this evaluation. We do not claim general superiority for the composite or treat these findings as limits on all geometric filtering methods.

Our first contribution is an adversary-free comparison at Dirichlet concentrations $\alpha \in \{0.5, 0.1, 0.01\}$. We state the dataset coverage explicitly and measure honest-client exclusion separately from changes in final accuracy. Fixed-cardinality selection gives an exact exclusion rate. Majority-cluster retention gives an upper bound, whose attainment we assess by directly counting the saved FLAME acceptance sets rather than inferring their sizes from rounded rejection rates. Random subsets that keep a rule’s admitted count in every round then separate how many updates it admits from which ones.

Second, we give a conditional analysis of the composite’s decision path. Set containment identifies which initial exclusions later stages cannot undo, and convexity gives a sufficient condition under which centroid validation rejects

S. G. Özden is with the Department of Computer Engineering, Faculty of Engineering and Architecture, Tokat Gaziosmanpaşa University, Tokat, Türkiye (e-mail: sebahattin.ozden0025@gop.edu.tr; ORCID: 0009-0008-9280-444X).

K. Sarıkaya is with the Department of Industrial Engineering, Faculty of Engineering and Architecture, Tokat Gaziosmanpaşa University, Tokat, Türkiye (ORCID: 0000-0001-9015-6209).

Corresponding author: Sebahattin Gökçen Özden.

no group when every candidate group lies inside the initial acceptance pool. We check that condition on recorded checkpoints, including re-executed canonical runs, and report where its assumptions hold. A separate score-ordering result states when a rule rejects honest clients but no attackers. This result requires an explicit ordering of scores; a distance constraint on the attack alone does not guarantee that ordering for every defense score.

Third, fixed-matrix replays test two changes to the treatment of unclustered low-density updates. Testing them as singletons adds no attacker rejections on the examined matrices, whereas excluding them outright gives a different trade-off in one patch-backdoor block. The replays separate a lack of test coverage from a failure of the validation rule itself. They concern decisions at recorded states, not the accuracy or attack success of retrained variants.

The evaluation comprises 2,130 planned rule-condition-seed units, of which 2,125 completed; the five failures remain in the accounting. This retrospective, descriptive analysis connects controlled observations to explicit decision rules and sufficient conditions. It establishes neither new universal lower bounds nor prospective confirmation of a theory.

Sections II–V cover related work, the model, the conditional analysis, and the protocol; Sections VI–XI report and discuss the results.

II. RELATED WORK

A. Robust aggregation and heterogeneous data

Neighbour-based selection, coordinate-wise estimators, geometric-median aggregation, and trust bootstrapping limit the influence of poisoned updates under different assumptions [5], [6], [17], [7]. For heterogeneous data, prior analyses establish statistical limits and give robust aggregation constructions under stated assumptions [18], [19]. Our results concern a different quantity, namely which clients are admitted or excluded in a given round. The elementary set and convexity arguments of Section IV neither replace bounds on optimization error nor imply a bound on model accuracy.

FLAME combines clustering, clipping, and noise [8]. DeepSight inspects model behaviour and combines clustering with suspicion estimates [20]; FLSHield relies on client-side validation [21]; CrowdGuard and MESAS use multiple stages or statistics [22], [23]; and Fedward and FLAC develop clustering-based defenses further [24], [25]. Density-based filtering is still being developed: Venkata Ramana *et al.* combine adaptive neighbourhood selection, multi-metric density estimation, and median-absolute-deviation thresholding, and report a lower false-positive rate than a fixed-bandwidth local-density baseline [26]. That comparison, like the others cited here, is measured with adversaries present. These designs use different observations and decision rules. Applying an algebraic property of one stage to another therefore requires checking its assumptions.

The composite studied here adapts relative-density partitioning and shared-neighbour clustering from MDBSCAN [27]. We assess a local adaptation that differs from the original clustering algorithm in its use as a filter, its neighbourhood

cutoff, and its treatment of unclustered points. The accepted-set inclusion argument is a familiar property of conjunctive cascades [28]; what matters here is whether the deployed fallback preserves inclusion in the initial pool. Work on centroid anomaly detection under attack provides related context for the interaction between acceptance regions and adversarial inputs [29].

B. False positives and prior evaluations

Fairness studies have examined how defenses exclude statistically unusual honest clients [11], [14], [15]. Recent evaluations of federated learning under non-IID data likewise document deteriorating defense performance [30], [12]. Bellachia *et al.* explicitly describe attackers hidden within a dispersed honest population [13], and we attribute that mechanism to their work.

Operating points below the chance diagonal also appear in earlier studies. Table 8 of Bellachia *et al.* reports $\text{TPR} = 0$ and $\text{FPR} = 0.45$ for FLAME under a non-IID IBA condition [13]. In Table V of Ye *et al.*, the reported TPR/TNR pairs at $\text{Dir}(0.1)$ imply $J = -4.15$ and -6.8 percentage points for FoolsGold and CrowdGuard, respectively [12]. Negative J is therefore not specific to the distance-constrained probes evaluated here.

Rates from other studies need care. Ye *et al.* report FLAME TNR values of 73.29% at $\text{Dir}(0.5)$ and 54.87% at $\text{Dir}(0.1)$, that is, FPR values of 26.71% and 45.13%, but on attacked rounds with other cohort sizes and denominators, so they neither confirm nor contradict the 48.89% upper bound of our configuration.

Evaluation methodology is itself a subject of study. Shejwalkar *et al.* examine how realistic poisoning threat models are [31], and Abad *et al.* discuss the role of controls without a backdoor [32]. A recent heterogeneity-aware survey likewise argues that defense evaluations should report false positives against benign non-IID clients and benign-client exclusion [16]. Our contribution is a reproducible combination of adversary-free controls, decision-path measurements, and conditional analysis of one composite. We claim priority neither for benign controls nor for the observation of negative J .

III. SYSTEM MODEL AND DECISION RULES

A. Protocol and threat model

At round r , client i receives w_r and submits the local model delta $u_{r,i} = w_{r,i}^{\text{local}} - w_r$. A uniform-mean filter that accepts the indices B_r applies $w_{r+1} = w_r + |B_r|^{-1} \sum_{i \in B_r} u_{r,i}$. The other rules in the panel use their own weighting, clipping, or coordinate-wise aggregation; uniform and sample-weighted averaging are included as separate baselines.

The server is trusted and client identities are fixed. Adversaries control their own training data or submitted updates. The Min-Max and Min-Sum probes are omniscient: they also observe the current benign updates and construct a common malicious vector subject to the respective distance bound. Ground-truth malicious identities are used to construct these

attacks and to score decisions; they are never an input to the defense.

Let $U = \{1, \dots, n\}$ index the submissions, and let H and A partition U into honest and adversarial participants. Assume $|H| > 0$. For accepted indices B , define $\text{FPR} = |H \setminus B|/|H|$ and, when $|A| > 0$, $\text{TPR} = |A \setminus B|/|A|$. Their difference is Youden's index $J = \text{TPR} - \text{FPR}$. In adversary-free rounds we call FPR the benign exclusion rate, BER. These quantities describe participation decisions, not accuracy or backdoor success.

Definition 1 (Score-order rule): Given real scores s_i , a score-order rule discards the k highest-scoring submissions or those with $s_i > \rho$ for a specified threshold. Ties require a specified selection convention.

Radial distance and nearest-neighbour distance sums are examples of such scores. The theorem in Section IV does not assume that different scores induce the same ordering. Cluster membership can be encoded as a binary score, but that encoding says nothing further about how the score relates to a distance-constrained attack.

Definition 2 (Monotone cascade): A cascade is monotone if each intermediate accepted set is a subset of its predecessor. A fallback that readmits previously removed indices can violate this property.

B. The evaluated composite

Fed-MDBSCAN-G was designed around a hypothesis about heterogeneous data. Under non-IID partitions, honest updates need not form a single dense cloud around the round's centre; they can form several groups of different densities. MDBSCAN [27] separates low-density points by relative density and clusters them with shared nearest neighbours, so it can in principle recover such groups instead of treating every sparse update as an outlier. A clustering stage alone, however, does not identify which group represents the benign consensus. We therefore anchor it at a geometric-median acceptance region, a reference with breakdown point one half that limits the influence of a minority of arbitrary updates [17]. Fig. 1 summarises the resulting three-stage decision path; the paragraphs below define each stage.

The initial filter of Fed-MDBSCAN-G accepts

$$B_0 = \{i : \|u_i - m\|_2 \leq \rho \text{median}_j \|u_j - m\|_2\}, \quad \rho = 2.5, \quad (1)$$

where m is a Weiszfeld approximation to the geometric median. If fewer than half of the submissions survive, the initial pool is reset to U .

For $k' = \min(5, n - 1)$, density is computed from the sum d_i of distances to the k' nearest neighbours. The local implementation returns $k'/\max(d_i, 10^{-15})$ when $d_i > 0$ and 1 when $d_i = 0$; a singleton input also returns 1. The deployed gate opens when the largest gap between consecutive sorted density values exceeds ten times their median gap and at least $\max(2, \lceil 0.05n \rceil)$ submissions are rejected initially. A three-round memory can keep the cluster stage active after an earlier activation, subject to decay over clean rounds.

When active, the cluster stage partitions the full submission set by density and runs shared-neighbour clustering on the low-density subset L . Each recovered group S is rejected if

$$\|c_S - m_{B_0}\|_2 > \tau R_0, \quad (2)$$

$$R_0 = \max(10^{-10}, \text{median}_{j \in B_0} \|u_j - m_{B_0}\|_2),$$

where c_S is the arithmetic centroid of the group, m_{B_0} is the geometric median of B_0 , and $\tau = 2$. Points that are not assigned to a recovered group are not tested at this stage. Because candidate groups are formed from L rather than from $L \cap B_0$, they can contain indices outside B_0 .

The combined set removes the members of rejected groups from B_0 . If fewer than half of the submissions then remain, a valve restores B_0 . The complete path is therefore not necessarily a monotone cascade, but its final output always satisfies $B \subseteq B_0$, which is all that the initial-exclusion result below requires.

IV. CONDITIONAL ANALYSIS

The statements in this section apply to a single round under the decision rules defined above. They use set inclusion, cardinality, and convexity. To explain an observed decision, their assumptions must hold along the implemented path; Section VIII checks these assumptions on recorded checkpoints.

A. Accepted-set containment and cardinality

Proposition 1 (Initial exclusion floor): If $B \subseteq B_0$, then $\text{FPR}(B) \geq \text{FPR}(B_0)$ and, for $|A| > 0$, $\text{TPR}(B) \geq \text{TPR}(B_0)$. This holds for a monotone cascade and for a fallback that restores at most B_0 .

Proof: The inclusions $H \setminus B_0 \subseteq H \setminus B$ and $A \setminus B_0 \subseteq A \setminus B$ give the inequalities after division by the corresponding population sizes. ■

Later stages therefore cannot readmit an honest client that the initial filter excluded. They can still improve the operating point, for instance by rejecting additional attackers without rejecting additional honest clients, so the proposition does not rule out a Pareto improvement.

Corollary 1 (Fixed-cardinality selection): If exactly m of n submissions are admitted, then $\text{BER} = 1 - m/n$ in an adversary-free round. With a adversaries and t rejected adversaries, $\text{FPR} = (n - m - t)/(n - a)$, provided $n > a$.

Proof: There are $n - m$ rejections, of which t are adversarial and $n - m - t$ are honest. ■

Our Multi-Krum configuration uses $f = \lceil 0.3n \rceil$ and $m = n - f - 2$. At $n = 90$ it retains 61 updates, so its clean BER is 29/90; with $a = 27$ and $t = 0$, its attacked-round FPR is 29/63. Agreement with these identities verifies selection cardinality only (Section V-D).

Corollary 2 (Majority-cluster retention): In an adversary-free round, a rule retaining exactly one cluster of size $c \geq c_{\min}$ has $\text{BER} = (n - c)/n \leq (n - c_{\min})/n$. Equality holds precisely when $c = c_{\min}$. With a adversaries and t adversarial rejections, $\text{FPR} = (n - c - t)/(n - a)$.

Proof: Count the rejected indices outside the cluster and subtract the adversarial rejections. ■

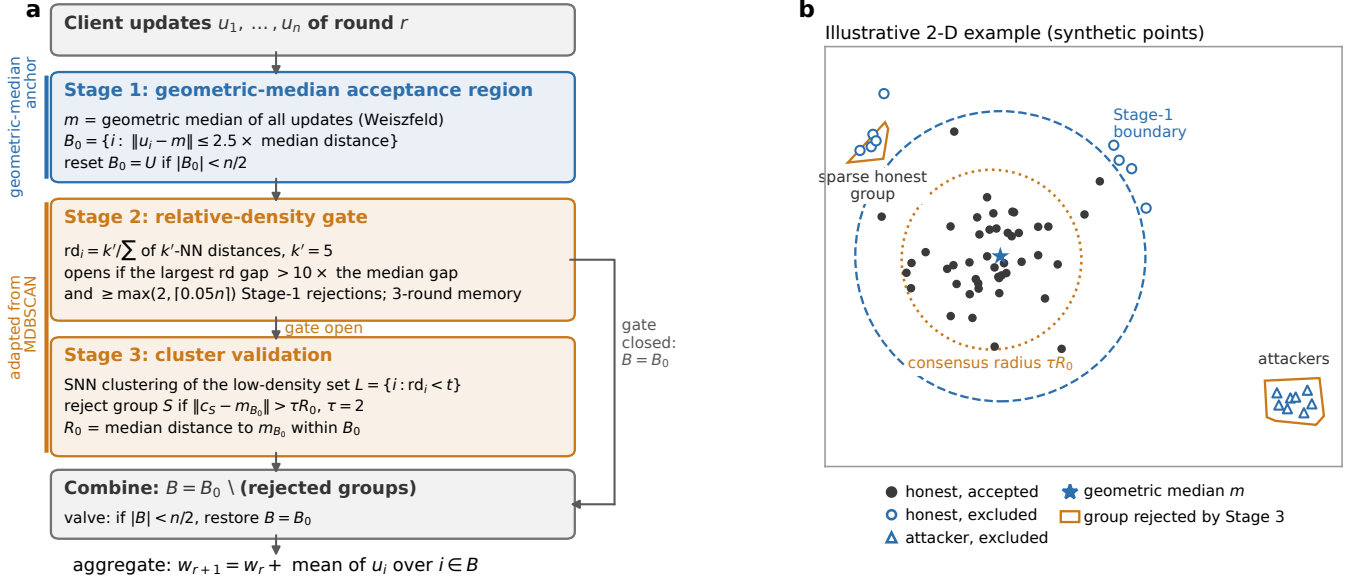


Fig. 1. Decision path of Fed-MDBSCAN-G. (a) Stage 1 forms the initial pool B_0 around the geometric median; Stage 2 opens the cluster stage when a relative-density gap coincides with Stage-1 rejections, and a three-round memory can keep it open; Stage 3 clusters the low-density set and rejects groups whose centroid lies farther than τR_0 from the geometric median of B_0 . The final set always satisfies $B \subseteq B_0$. (b) Illustrative two-dimensional example produced by the deployed stage functions on synthetic points (not experimental data). Stage 1 excludes the attackers together with nine honest points; the two groups rejected by Stage 3 lie entirely outside B_0 , so the cluster stage leaves the accepted set unchanged. The example shows the mechanics only: on the recorded attacked checkpoints of Section VIII, Stage 1 excludes at most a few attackers except under the patch attack at $\alpha = 0.01$, and that section reports how often each stage executed.

The local FLAME path uses $c_{\min} = \lfloor n/2 \rfloor + 1 = 46$ at $n = 90$, which gives a clean upper bound of $44/90$. Heterogeneity alone does not force equality, and rounds in which clustering fails or a fallback fires must be treated according to the path actually executed.

B. Centroid validation

Proposition 2 (Convex-hull radius bound): For a nonempty group S , any convex combination $g(S)$ of its updates, and any reference m_0 ,

$$\|g(S) - m_0\| \leq \max_{i \in S} \|u_i - m_0\|.$$

For the uniform mean of s members, if one member has radius at most r_a and the others at most r_b , then

$$\|c_S - m_0\| \leq \frac{r_a + (s-1)r_b}{s}.$$

Proof: Write $g(S) - m_0$ as a convex combination of the vectors $u_i - m_0$ and apply the triangle inequality. The second expression uses weights $1/s$. ■

If a group centroid lies outside a ball, at least one member must also lie outside. This does not mean that every member lies outside. Rejecting the whole group can therefore exclude members inside the ball: on the real line, with centre 0 and radius 2, the group $\{1, 5\}$ has centroid 3 and is rejected although its member 1 lies inside. Conversely, cancellation can place a centroid inside the ball even when a member lies outside. The initial and consensus filters also use different reference points and radii, so the bound does not order their client-level decisions.

Proposition 3 (Sufficient condition for inactive centroid validation): For nonempty B_0 , define

$$\Gamma(B_0) = \frac{\max_{i \in B_0} \|u_i - m_{B_0}\|_2}{R_0},$$

where R_0 is the floored radius in (2). If $\Gamma(B_0) \leq \tau$, then no nonempty group $S \subseteq B_0$ is rejected by (2). If every tested group is such a subset and no other stage removes members, the final accepted set is $B = B_0$.

Proof: Every member of B_0 lies in the closed ball of radius τR_0 about m_{B_0} . The ball is convex, so every convex combination of members of a subset $S \subseteq B_0$ lies in it, and the strict rejection inequality cannot hold. ■

The condition is sufficient but not necessary, and it does not hold in general for our composite, whose candidate groups can include members outside B_0 . For example, $B_0 = \{-1, 1\}$ has $\Gamma = 1$, yet the group $\{1, 10\}$ has centroid 5.5 and fails the radius-2 test. Before applying the proposition to recorded matrices, we therefore verify the subset condition.

C. Density ordering and its implementation boundary

Proposition 4 (Density separation with a separating threshold): Consider the ideal score $rd(i) = k'/d_i$ with strictly positive distance sums. Suppose $|A| \geq k' + 1$, all adversarial pairwise distances are at most $\delta_A > 0$, and each honest point has mean k' -nearest-neighbour distance at least $\delta_H > \delta_A$. Then

$$\min_{a \in A} rd(a) \geq 1/\delta_A > 1/\delta_H \geq \max_{i \in H} rd(i).$$

For a threshold satisfying $\max_{i \in H} \text{rd}(i) < t \leq \min_{a \in A} \text{rd}(a)$, the set $L = \{i : \text{rd}(i) < t\}$ contains every honest point and no adversarial point.

Proof: Each attacker has k' other attackers within δ_A , so its nearest-neighbour distance sum is at most $k'\delta_A$. The assumed honest mean distance gives $d_i \geq k'\delta_H$. Taking reciprocals yields the ordering, and the threshold condition gives the selected set. ■

The restriction on the threshold is essential, because a threshold above every score selects attackers as well. The implementation also has a boundary case: when the effective nearest-neighbour distances sum to exactly zero, the code returns $\text{rd} = 1$; for the one-dimensional input $[0, 0, 0.1, 0.2]$ with $k' = 1$, the actual output is $[1, 1, 10, 10]$. On the recorded high-dimensional matrices, however, the nearest-neighbour search returns small positive distances even between bit-identical updates, so this branch is never taken there (Section VIII). The clustering intuition agrees with earlier observations [13], but explaining a particular run requires its computed scores and routing decisions.

D. A conditional score-ordering result

Theorem 1 (Honest-only rejection): Let H, A both be nonempty and $1 \leq k \leq |H|$. If a rule discards the k highest scores and $\max_{a \in A} s_a < s_{(k)}^H$, where $s_{(k)}^H$ is the k th largest honest score, then

$$\text{TPR} = 0, \quad \text{FPR} = k/|H|, \quad J = -k/|H|.$$

For a threshold rule, the same conclusion holds with $k = |\{i \in H : s_i > \rho\}| > 0$ if $\rho \geq \max_{a \in A} s_a$.

Proof: The strict ordering places all k discarded indices in H . In the threshold case, no attacker exceeds the threshold. Count honest and adversarial rejections. ■

The theorem is a conditional counting result. Min-Max and Min-Sum constrain distances [10], but those constraints do not establish its ordering premise for every score. For instance, the honest points $\{-2, 0, 0.1, 2\}$ have maximum pairwise distance 4, and the point -1 lies within distance 3 of each of them; yet a rule that rejects the three points farthest from the honest geometric median rejects -1 together with -2 and 2 . Negative J also says nothing directly about model accuracy or attack success, and a larger rejection budget makes J more negative only while the honest-only ordering continues to hold.

Neither the containment nor the convexity result depends on how the reference point is obtained, so both also apply to rules that use trusted server data. The positive family-average J that FLTrust attains in Section VII is an empirical observation; it does not show that an external reference is necessary or sufficient for robustness.

V. EXPERIMENTAL PROTOCOL

A. Simulation and records

All experiments run in a federated simulation written in PyTorch: in each round, the scheduled clients are trained one after another in a single process, and their updates are passed to the server-side rule. Communication, latency, and device compute are not modelled. The setting reflects

TABLE I
PLANNED EVALUATION COVERAGE BY BLOCK

Block	Units
Main matrix (attacked and adversary-free)	1500
Ablation variants	252
Matched attack-disabled controls	156
Oracle diagnostics	78
Alternative partition policy	54
Fixed local-step control	54
Onset/offset timing	36
Total planned	2130

A unit is one rule–condition–seed run. Of 2130 planned units, 2125 completed and 5 are retained as failures rather than imputed or re-seeded. The blocks are not a balanced full factorial.

TABLE II
EXPERIMENTAL SETUP

Item	Setting
MNIST, Fashion-MNIST	multilayer perceptron 784–200–10
UCI HAR	multilayer perceptron 561–128–64–6, dropout 0.2
CIFAR-10	two convolutional layers (32, 64 channels) and fully connected layers 256–10
Clients	100; 90 per round, adversaries in every round
Training	30 rounds, 3 local epochs, batch size 32
Optimizer	learning rate 0.01, momentum 0.9
Seeds	42, 137, 2024
Partition	Dirichlet, $\alpha \in \{0.5, 0.1, 0.01\}$; lognormal retention, $\sigma = 0.5$; ≥ 20 examples per client
Root set	100 examples, held out for all rules
Software	Python 3.12, PyTorch 2.11, CUDA

cross-device and Internet-of-Things deployments through its data: non-IID partitions, unequal client dataset sizes, partial participation, and a smartphone-sensor dataset (UCI HAR). PyTorch’s deterministic mode was off, so GPU results need not repeat bit for bit. Each run records, per round, the scheduled, participating, accepted, and rejected identities, the resulting confusion counts, test accuracy, and patch-attack success, with hashes of its partition, schedule, and initial model. The tables in Sections VI and VII are computed from these records.

B. Data, training, and coverage

Table II summarises the datasets [33], [34], [35], [36], models, and training settings. A Dirichlet draw assigns training examples to clients [37]. Each client then keeps a random fraction $\min(1, e^Z)$, $Z \sim \mathcal{N}(0, 0.5^2)$, of its share, but at least 20 examples where available; discarded examples are not reassigned. Next, 100 examples drawn uniformly from all client data are held out as the FLTrust root set for every rule, so all rules see the same client data for a given seed. Finally, a minimum-count repair moves examples from the largest clients until each holds at least 20. A seeded schedule selects 90 of the 100 clients per round; adversaries have fixed identities and join every round, whereas each honest client joins a random subset of rounds. MNIST, Fashion-MNIST, and UCI HAR share a 17-condition main block, whereas CIFAR-10 has smaller coverage. The archived evaluation comprises 149 jobs and 2,130 planned rule–condition–seed units (Table I), of which 2,125 completed. The five failures, four sample-weighted HAR runs and one CIFAR FLAME run under Min-Max, are kept in

the accounting and are neither imputed nor replaced. For this revision, 39 canonical runs were re-executed and 36 control runs were added; they are reported separately and leave the archived totals unchanged (supplement).

C. Control blocks and attacks

The adversary-free main conditions configure zero malicious clients at each of the three concentrations; CIFAR has no $\alpha = 0.5$ control cell. For the patch-backdoor comparison, attack-disabled controls keep the client identities and schedules of the attacked configurations and switch the attack off. The oracle, alternative-partition, fixed-step, and timing blocks of Table I are diagnostics outside all reported tables: oracle runs drop the active adversaries before aggregation, the alternative partition skips the repair, the fixed-step control caps local training at five optimizer steps, and the timing block attacks only in rounds 10 to 19.

The five attack families are loud Gaussian noise (coordinate standard deviation 5), label flipping, a bottom-right 3×3 white-patch backdoor (target class 0, poisoning fraction 0.2), bounded random perturbations, and constrained omniscient probes. The bounded family comprises a direction probe at relative scale 0.005 and random noise at relative scale 0.3; the latter carries the legacy code name `adaptive_gaussian` but is not an optimized adaptive attack. The constrained probes submit a common vector opposite the benign mean, scaled to satisfy a Min-Max or Min-Sum bound [10].

D. Implementations

The panel comprises the uniform mean, the sample-weighted mean, norm clipping, the coordinate-wise median, Multi-Krum with $f = \lceil 0.3n \rceil$ and $m = n - f - 2$, normalized FLTrust with 100 root examples, a local HDBSCAN-based FLAME with noise level 0.001, and Fed-MDBSCAN-G, together with seven ablation variants. We checked the local FLTrust rule against a transcription of the authors' demonstration code on 18 recorded matrices, with relative discrepancy below 2.8×10^{-7} . On six of the five-step Min-Max checkpoint matrices of Section VIII, the local Multi-Krum scores match the published rule [5], and the selections differ only in how ties among identical attacker copies are broken, which leaves the aggregate unchanged to float32 precision. We also exercised FLAME's clipping, averaging, and noise scaling through the server's aggregation path. For all 200 completed FLAME runs (6,000 rounds) we counted the saved accepted identities, which gives the retained-set sizes used in Section VI. The scope of these checks is detailed in the supplement and summarised in Section X.

E. Metrics

Benign utility cost is uniform-mean final accuracy minus defense final accuracy in the same adversary-free condition; under attack, utility benefit is defense accuracy minus uniform-mean accuracy. Accuracy comparisons use the completed seeds of each cell. We also report attacker recall, honest rejection, J , and the attack success rate (ASR) on triggered non-target examples. FLTrust assigns zero weight to submissions

with non-positive trust, and its BER, FPR, and TPR count these zero-weight decisions. Alarm specificity is reported only for the composite, the one rule in the panel that exposes a round-level alarm.

VI. ADVERSARY-FREE EXCLUSION AND UTILITY

Without any attacker, the rejecting rules exclude a large share of honest submissions in many conditions (Table III, Fig. 2). Multi-Krum excludes exactly 32.22% in all nine available cells, as Corollary 1 requires.

FLAME's clean exclusion is bounded by its cluster-size rule, and in most clean rounds it reaches that bound. It keeps exactly the smallest admissible majority, 46 of 90 updates, in 719 of 990 rounds (Table V), and its cell-level exclusion lies between 42.4% and 48.9%. Under strong heterogeneity, most visibly on HAR, it retains larger clusters and reaches the bound less often, so the realized exclusion also depends on the data.

Fed-MDBSCAN-G behaves differently. Its exclusion stays below 2.1% at $\alpha = 0.5$ but reaches 42.6% at $\alpha = 0.01$ (Table III), because its Stage-1 threshold scales with each round's median radius and the accepted fraction follows the shape of that distribution.

Exclusion is only one source of utility loss (Table IV). The coordinate-wise median rejects no client, yet on MNIST at $\alpha = 0.01$ its final accuracy is 64.4 percentage points below uniform averaging; the largest loss, 65.55 points, is FLAME's on the same condition. Averaged over the reported cells, the loss grows from 3.3 to 27.5 points as α decreases from 0.5 to 0.01, so admission rates alone are insufficient to judge benign utility. For Multi-Krum at $\alpha = 0.01$, which updates are kept matters more than how many (Table VI): replacing its 61 selected updates with a random subset of the same size raises final accuracy by 15.9 to 43.7 points on average, to within 1.3 points of uniform averaging on MNIST and Fashion-MNIST. For FLAME, a random subset matched to the canonical admitted count in every round raises accuracy by 7.2 to 24.6 points, in every seed pair, whereas removing its noise changes accuracy by at most 0.33 points. The remaining gap of 8.0 to 41.0 points to uniform averaging reflects FLAME's clipping, its smaller admitted set, or both.

The composite's round-level alarm has poor specificity under strong heterogeneity (Table VII): at $\alpha = 0.01$ it fires in every adversary-free round on every dataset, and at $\alpha = 0.1$ in almost every HAR round. All of these alarms are false.

VII. DISCRIMINATION UNDER ATTACK

Attack discrimination depends strongly on the attack family (Table VIII; per rule in the supplement). Averaged over the four rejecting rules, J falls from +68.6 points for loud Gaussian noise, through label flipping, to near zero for patch backdoors and bounded perturbations, and becomes negative (−16.1) for the constrained probes. Because the families also differ in adversary ratio, concentration, and dataset coverage, this ordering is descriptive.

Under the constrained probes, Multi-Krum rejects honest clients only in five of its six cells (Table IX). In each of

TABLE III
BENIGN EXCLUSION RATE (BER, %) WITHOUT ADVERSARIES

Rule	$\alpha = 0.5$			$\alpha = 0.1$				$\alpha = 0.01$			
	MNIST	Fashion	HAR	MNIST	Fashion	HAR	CIFAR-10	MNIST	Fashion	HAR	CIFAR-10
Uniform mean (undefended)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sample-weighted mean (undefended)	0.0	0.0	0.0	0.0	0.0	0.0	–	0.0	0.0	0.0	–
Norm clipping	0.0	0.0	0.0	0.0	0.0	0.0	–	0.0	0.0	0.0	–
Coordinate-wise median	0.0	0.0	0.0	0.0	0.0	0.0	–	0.0	0.0	0.0	–
Multi-Krum	32.2	32.2	32.2	32.2	32.2	32.2	–	32.2	32.2	32.2	–
FLTrust (normalized)	33.1	31.9	11.9	25.3	27.3	19.6	–	30.3	32.3	31.0	–
FLAME (HDBSCAN)	48.9	48.9	48.3	48.9	48.7	47.4	48.7	48.7	47.0	42.4	45.5
Fed-MDBSCAN-G (full)	0.3	0.4	2.1	0.2	0.5	24.1	0.1	42.6	38.7	24.4	39.9

Honest client-round submissions rejected with no adversary present; three seeds, 30 rounds and 90 participants per round (8,100 honest client-rounds per cell). Fashion denotes Fashion-MNIST; dashes mark conditions outside the planned coverage. FLTrust counts zero-weight submissions. Norm clipping and the coordinate-wise median reject no client, so their BER is zero by construction; their benign cost appears in Table IV.

TABLE IV
BENIGN UTILITY COST (BUC, PERCENTAGE POINTS) WITHOUT ADVERSARIES

Rule	$\alpha = 0.5$			$\alpha = 0.1$				$\alpha = 0.01$			
	MNIST	Fashion	HAR	MNIST	Fashion	HAR	CIFAR-10	MNIST	Fashion	HAR	CIFAR-10
Norm clipping	0.0	0.0	+0.2	0.0	0.0	+2.2	–	+22.5	+13.3	0.0	–
Coordinate-wise median	+0.6	+0.8	+9.8	+4.7	+5.1	+33.3	–	+64.4	+40.9	+23.8	–
Multi-Krum	+1.0	+1.3	+2.8	+4.6	+4.8	+29.2	–	+44.9	+33.6	+11.0	–
FLTrust (normalized)	+2.4	+2.5	+13.2	+3.3	+4.1	+29.2	–	+11.6	+5.1	+10.6	–
FLAME (HDBSCAN)	+1.7	+1.9	+19.6	+10.4	+6.7	+38.1	+15.5	+65.5	+45.9	+15.2	+15.6
Fed-MDBSCAN-G (full)	0.0	0.0	+1.8	0.0	+0.1	+26.8	0.0	+59.2	+41.9	+10.4	+14.9

Final-round accuracy of the undefended uniform mean minus that of the defense, in the runs of Table III; positive values denote lower accuracy than the uniform mean. Means over three seeds; per-seed ranges are given in the supplement.

TABLE V
FLAME RETAINED CLUSTER SIZES WITHOUT ADVERSARIES

Dataset	α	Mean size	Range	Size 46
MNIST	0.5	46.01	46–47	89/90
MNIST	0.1	46.03	46–47	87/90
MNIST	0.01	46.17	46–49	77/90
Fashion	0.5	46.01	46–47	89/90
Fashion	0.1	46.19	46–50	81/90
Fashion	0.01	47.73	46–56	39/90
HAR	0.5	46.50	46–53	67/90
HAR	0.1	47.30	46–51	38/90
HAR	0.01	51.83	46–62	9/90
CIFAR-10	0.1	46.13	46–48	79/90
CIFAR-10	0.01	49.09	46–61	64/90

Counted from saved accepted identities in the adversary-free main cells; each row covers three seeds and 90 rounds. Size 46 is the minimum admissible size, $\lfloor n/2 \rfloor + 1$.

TABLE VI
ADVERSARY-FREE FINAL ACCURACY (%) AT $\alpha = 0.01$: RULES VERSUS RANDOM SUBSETS OF THE SAME SIZE

Dataset	FedAvg	Multi-Krum rule	Multi-Krum random	FLAME rule	FLAME random
MNIST	85.65	40.71	84.36	20.10	44.65
Fashion-MNIST	73.63	40.04	72.58	27.74	41.86
UCI HAR	45.53	34.57	50.44	30.35	37.50

Means over three seeds. Random subsets replace a rule's selected updates in every round and keep its admitted count: 61 for Multi-Krum and, for FLAME, the count that the canonical FLAME run recorded in that round. Clipping and noise are unchanged; paired differences are in the supplement.

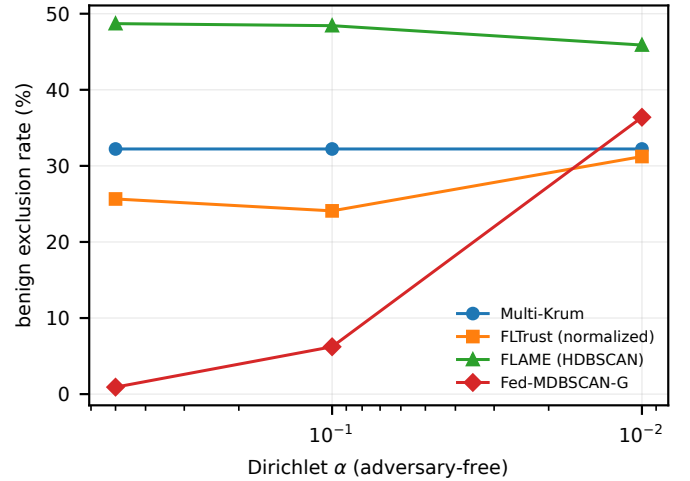


Fig. 2. Adversary-free exclusion, averaged over available dataset cells. Coverage varies by concentration and method. FLTrust counts zero-weight submissions. The FLAME bound need not be attained in every round; direct sizes appear in Table V.

these cells, all 2,610 rejections are honest, giving $FPR = 29/63 = 46.03\%$, the value that Corollary 1 predicts at zero attacker recall. The exception is HAR at $\alpha = 0.1$, where attacker recall is positive ($J = +36.68$ points). Because all constrained attackers submit the same vector, zero recall could also arise from tie-breaking at the selection boundary, so these counts do not by themselves establish the ordering premise of Theorem 1.

TABLE VII
BENIGN ROUND-LEVEL ALARM RATE OF FED-MDBSCAN-G

α	Dataset	Alarming rounds	Rate (%)	BER (%)
0.5	MNIST	0/90	0.0	0.27
0.5	Fashion	0/90	0.0	0.41
0.5	HAR	19/90	21.1	2.07
0.1	MNIST	0/90	0.0	0.19
0.1	Fashion	0/90	0.0	0.53
0.1	HAR	89/90	98.9	24.06
0.1	CIFAR-10	0/90	0.0	0.12
0.01	MNIST	90/90	100.0	42.57
0.01	Fashion	90/90	100.0	38.69
0.01	HAR	90/90	100.0	24.37
0.01	CIFAR-10	90/90	100.0	39.94

The denominator is adversary-free rounds, so every alarm is false. Fed-MDBSCAN-G is the only evaluated rule with a round-level alarm; the other rules have no alarm channel, and no rate is reported for them.

TABLE VIII
CLIENT-LEVEL DISCRIMINATION BY ATTACK FAMILY

Attacker family	Cells	TPR (%)	FPR (%)	J (pp)	Benefit (pp)
Loud Gaussian ($\sigma = 5$)	60	87.5	18.9	+68.6	+13.7
Label flipping	36	69.7	24.8	+45.0	-9.0
Patch backdoor	16	34.0	32.3	+1.8	-19.2
Bounded random direction	72	34.1	33.7	+0.4	-23.0
Constrained omniscient	28	21.8	38.0	-16.1	-10.6

$J = \text{TPR} - \text{FPR}$ (Youden's index); $J < 0$ means honest clients are rejected in preference to attackers. Benefit is defense accuracy minus undefended accuracy in the same condition. Each row pools rule $\times$ dataset $\times$ condition cells of the four rejecting rules; family coverage differs, so the rows are descriptive averages.

TABLE IX
OPERATING POINTS UNDER THE CONSTRAINED OMNISCIENT ADVERSARY

Dataset	Rule	TPR (%)	FPR (%)	J (pp)	Predicted FPR (%)
$\alpha = 0.1$					
MNIST	Multi-Krum	0.00	46.03	-46.03	46.03
MNIST	FLAME (HDBSCAN)	0.00	68.11	-68.11	≤ 69.84
Fashion	Multi-Krum	0.00	46.03	-46.03	46.03
Fashion	FLAME (HDBSCAN)	0.00	65.59	-65.59	≤ 69.84
HAR	Multi-Krum	57.90	21.22	+36.68	46.03
HAR	FLAME (HDBSCAN)	96.67	27.25	+69.42	≤ 69.84
CIFAR-10	FLAME (HDBSCAN)	8.33	62.94	-54.60	≤ 69.84
$\alpha = 0.01$					
MNIST	Multi-Krum	0.00	46.03	-46.03	46.03
MNIST	FLAME (HDBSCAN)	0.00	58.82	-58.82	≤ 69.84
Fashion	Multi-Krum	0.00	46.03	-46.03	46.03
Fashion	FLAME (HDBSCAN)	0.00	63.19	-63.19	≤ 69.84
HAR	Multi-Krum	0.00	46.03	-46.03	46.03
HAR	FLAME (HDBSCAN)	0.00	59.47	-59.47	≤ 69.84
CIFAR-10	FLAME (HDBSCAN)	0.00	57.51	-57.51	≤ 69.84

Thirty percent adversaries: 63 honest and 27 adversarial participants per round. Predicted FPR is the value at zero attacker recall (TPR = 0): an identity for Multi-Krum and an upper bound for FLAME (Corollaries 1–2). FLTrust and Fed-MDBSCAN-G yield no such prediction; their per-rule rates are in the supplement. Each cell has 90 recorded rounds except CIFAR-10/FLAME/ $\alpha = 0.1$ (60 rounds from two completed seeds).

FLAME behaves similarly. It has zero attacker recall in six of its eight constrained-probe cells, with FPR between 57.51% and 68.11%, below the $44/63 = 69.84\%$ ceiling that its retained majority imposes when no adversary is rejected. Fed-

TABLE X
PATCH-BACKDOOR RESULTS OF FED-MDBSCAN-G WITH MATCHED CONTROLS

α	Dataset	Attacked		Attack disabled	
		Acc. (%)	ASR (%)	Acc. (%)	ASR (%)
0.1	MNIST	93.97	99.99	94.14	0.57
0.1	Fashion	83.34	99.93	83.42	2.63
0.1	CIFAR-10	49.55	96.59	49.85	5.76
0.01	MNIST	28.20	79.85	26.01	5.52
0.01	Fashion	32.83	56.18	32.24	1.85
0.01	CIFAR-10	15.47	26.86	15.83	20.80

ASR is the triggered non-target attack success rate. Accuracy and ASR are reported together; the low-accuracy $\alpha = 0.01$ block should not be read as effective protection.

MDSCAN-G also has low or zero attacker recall in several of these cells.

FLTrust is the only rule with a positive mean J for the constrained family (+35.6 points), which points to trusted-data signals as a direction for further study. It also assigns zero weight to 11.9–33.1% of honest submissions in clean rounds (Table III).

Low attack success does not by itself indicate protection (Table X). At $\alpha = 0.1$ the composite keeps near-control accuracy, yet the backdoor succeeds on more than 96% of triggered examples on all three datasets. At $\alpha = 0.01$ attack success is lower, but main-task accuracy is at most 32.8%, and even the unpoisoned control assigns the target label to 20.8% of patched CIFAR examples. Accuracy, attacked ASR, and control ASR therefore have to be read together.

VIII. DECISION-PATH ANALYSIS

This section asks why the stages above Stage 1 change final accuracy so little. The canonical round records answer part of the question (supplement). In adversary-free, constrained-probe, and patch runs, the deployed gate opens in every round at $\alpha = 0.01$ but seldom at larger α outside HAR. When it opens under the constrained probes, the cluster stage never removes a Stage-1-accepted update, whereas in adversary-free and patch rounds at $\alpha = 0.01$ it removes some in 67 of 360 and 89 of 270 rounds. The later stages thus change accepted sets under strong heterogeneity, yet these changes barely move final accuracy.

We analyse two sets of recorded matrices, each under two gate settings and two neighbourhood-cutoff choices. The first holds 72 matrices from 24 Fed-MDBSCAN-G runs of a separate study of the neighbourhood cutoff: MNIST and Fashion-MNIST, three seeds, and four blocks of 18, namely Min-Max at $\alpha = 0.01$ and patch at $\alpha = 0.1$, each with an attack-disabled control. These runs cap local training at five optimizer steps per round and enable PyTorch's deterministic mode, and their attacked runs have 18 rather than 27 adversaries among 90 participants. The second set re-executes 30 canonical runs on MNIST and Fashion-MNIST with three seeds: Min-Max at $\alpha = 0.1$, Min-Sum at $\alpha = 0.01$, patch at $\alpha = 0.1$ and 0.01 , and adversary-free at $\alpha = 0.01$. In both sets every re-executed run matched all non-timing fields of its 30 archived round

TABLE XI
RADIAL RATIO OF THE INITIAL ACCEPTANCE POOL ON ATTACKED CHECKPOINTS

Attack	Matrices	$\Gamma \leq 2$	Median	Maximum
<i>Five-step development runs</i>				
Min-Max, $\alpha = 0.01$	18	18	1.257	1.622
Patch, $\alpha = 0.1$	18	17	1.666	2.017
All attacked	36	35	1.448	2.017
<i>Canonical runs</i>				
Min-Max, $\alpha = 0.1$	18	10	1.956	2.355
Min-Sum, $\alpha = 0.01$	18	3	2.429	5.360
Patch, $\alpha = 0.1$	18	9	2.023	2.415
Patch, $\alpha = 0.01$	18	0	5.520	13.024
All attacked	72	22	2.248	13.024

Distinct attacked checkpoint matrices; the threshold is $\tau = 2$. Five-step matrices come from the separate cutoff study with 18 of 90 adversaries, canonical ones from re-executed canonical runs with 27 of 90. In every five-step matrix B_0 contains all 90 participants; the five-step exception is Fashion, seed 137, patch, round 29.

records, and updates were saved at rounds 0, 9, and 29, so the second set holds the canonical runs' own matrices.

A. Gate execution and the measured radius condition

On the five-step matrices, the deployed gate runs the cluster stage on 10 of the 72 checkpoints and on none of the 36 attacked ones. A density-only gate, which drops the requirement for supporting Stage-1 rejections, leaves the accepted set unchanged in all 144 comparisons.

Table XI reports the radial ratio $\Gamma(B_0)$, computed with the floored radius in (2). In every attacked five-step matrix B_0 contains all 90 updates, so every candidate group lies inside B_0 , and the sufficient condition holds on **35 of 36 attacked matrices**: all 18 Min-Max and 17 of 18 patch matrices. There, regrouping members of B_0 cannot move a centroid outside radius $2R_0$, so the consensus test rejects no group even when the gate is opened; at the remaining matrix the recovered groups are still accepted. The canonical matrices behave differently: the condition holds on 22 of their 72 attacked matrices and on none of the patch matrices at $\alpha = 0.01$, where the median Γ is 5.5. The radius argument therefore explains the five-step checkpoints but only part of the canonical ones.

B. Routing and validation are different events

Opening the gate lets patch attackers reach the consensus test, but on the five-step matrices they pass it: under the density-only gate with the cutoff enabled, 239 of the 306 adversarial observations enter recovered groups and 30 of the 55 groups contain adversaries, yet none is rejected. For the constrained probes, routing alone prevents a rejection. In all 54 constrained matrices of both sets the attacker updates are bit-identical, but the nearest-neighbour search returns small positive distances between them, so they receive relative densities above 4×10^6 against at most 98 for honest updates. As Proposition 4 predicts, the partition places every attacker on the high-density side and every honest update in L , so the cluster stage tests only honest updates; on the canonical matrices it recovers no candidate group at all. Strong heterogeneity

TABLE XII
ENDPOINT ABLATIONS: VARIANT MINUS FULL FINAL ACCURACY (PERCENTAGE POINTS)

Variant	Cells	Mean	Min.	Max.
Fed-MDBSCAN-G, stage 1 only	12	-0.081	-1.50	+0.84
Geometric-median distance control	12	-0.081	-1.50	+0.84
Fed-MDBSCAN-G, no activation memory	12	0.000	0.00	0.00
Fed-MDBSCAN-G, no valve	12	+0.278	-0.27	+3.03
Trust-region multiplier 1.5	12	-0.916	-8.42	+2.90
Trust-region multiplier 2.0	12	-0.860	-3.09	+2.95
Trust-region multiplier 3.0	12	+0.575	-6.38	+12.82

Variant minus full composite, final accuracy in percentage points; cells are dataset $\times$ condition means over three seeds in the matched ablation block. Ranges describe heterogeneity across conditions, not confidence intervals.

changes the picture: on the canonical adversary-free and patch matrices at $\alpha = 0.01$, the test rejects groups in 17 of 36. In 9 of them the rejected groups lie outside B_0 ; in the other 8 the test removes 34 honest and 13 adversarial Stage-1-accepted updates.

C. Interventions on unclustered updates

Testing unclustered updates as singletons adds honest rather than attacker rejections, and excluding them outright gives a positive J in only one block (supplement). Holding geometry, history, and group membership fixed, we compare the deployed policy, which tests recovered groups only (P0), with testing unclustered low-density updates as singletons (P1) and excluding them outright (P2). P1 adds no attacker rejections under either gate, but 32 honest rejections under the deployed gate and 77 under the density-only gate. P2 gives the only positive trade-off, in the density-only patch block ($J = +3.24$ points), at the cost of 82 honest rejections in the matching control.

A clean MNIST checkpoint from a separate five-step run at $\alpha = 0.01$ shows the converse (supplement): in each of three mini-batch regimes, the largest recovered group lies at 0.12–0.13 of the rejection radius and every smaller group at 1.70–2.63 times it, so the test rejects all smaller groups although every participant is honest.

D. Endpoint ablations

Removing every stage above Stage 1 changes mean final accuracy by less than 0.1 percentage points (Table XII). Across 12 matched dataset-condition cells and 36 endpoint runs, the Stage-1-only variant differs from the full composite by -0.081 points on average, with cell differences between -1.50 and $+0.84$ points; removing the valve changes the mean by $+0.278$ points, and the variant without activation memory matches all 36 endpoints.

IX. DISCUSSION

Without an attacker, the cost to honest clients is substantial in several configurations. For fixed-cardinality selection it is determined by the selected count; cluster retention imposes a bound, and the direct counts show how often it is attained. Neither exclusion metric substitutes for a utility measurement:

aggregation can reduce accuracy without rejecting any client, and a model can retain its accuracy while learning a backdoor. At $\alpha = 0.01$, the random-subset controls show that the loss also depends on which updates are excluded: a random 61 of 90 costs little accuracy, whereas Multi-Krum’s geometric choice of the same number costs much more.

For Fed-MDBSCAN-G, the decision-path measurements give two design lessons. When every candidate group lies in an acceptance pool whose maximum radius is below the centroid-test threshold, no regrouping can make the test reject a group. And a density partition that sends tightly copied updates to its high-density side keeps exactly those updates away from the test. Both properties should be checked before clustering complexity is added; a different group statistic, reference point, threshold, or routing rule may behave differently.

A useful evaluation should report clean exclusion and utility at the intended heterogeneity levels, the exact admission and fallback rules, and attack-side TPR, FPR, accuracy, and ASR with matched controls. Reported together, these quantities expose failures that any one of them can conceal. Exclusion in a single round is a participation decision, not permanent client removal or a direct measure of harm to the model.

X. LIMITATIONS

Scope of the evidence. The results rest on three seeds, 30-round horizons, small models, and uneven coverage, and the lognormal size retention and the partition repair alter the sampled heterogeneity. Client-rounds are correlated rather than independent replicates, so all results are descriptive: we make no claim of statistical significance, equivalence, or general superiority for any defense. The attack families are specified constructions rather than an exhaustive search, and their labels do not encode a measured displacement.

Implementation fidelity. The baselines are local implementations. The FLTrust check covers the aggregation rule but not degenerate-root paths or the full training protocol. The Multi-Krum check covers the selection rule on six replay checkpoints, not the canonical training runs, whose update matrices were not saved. FLAME’s clustering selects the same updates as the reference `hdbscan` library in all 270 re-executed rounds we compared, but its noise calibration and the authors’ training protocol remain unverified; removing the noise changes adversary-free accuracy at $\alpha = 0.01$ by at most 0.33 points. Agreement with a cardinality identity shows that a rule keeps the right number of updates, not that it keeps the same updates as the original. Details are given in the supplement.

Retrospective analysis. The conditional arguments of Section IV and the checkpoint analyses were developed after the results had been inspected; content hashes identify the evidence files but do not establish preregistration. The replays use recorded trajectories, whereas a modified defense would change subsequent inputs, and CIFAR repeatability was established for a single diagnostic condition. The radius condition was established on five-step matrices and holds less often on canonical ones, and the canonical checkpoints cover two datasets and three rounds per run. The elementary identities

therefore explain the evaluated decision paths; they are not general impossibility theorems. Whether trusted data, temporal information, alternative update representations, or client-side validation would change these outcomes remains open.

XI. CONCLUSION

We related adversary-free exclusion and attack-side decisions to the selection behaviour of eight server-side rules in a federated-learning evaluation with 2,125 completed runs and five retained failures. Multi-Krum’s clean exclusion follows from its fixed cardinality, and direct FLAME counts show when its cluster-size bound is attained. For Fed-MDBSCAN-G, the verified subset condition and measured radius ratio explain why centroid validation cannot reject any candidate group on 35 of 36 attacked checkpoint matrices from separate five-step runs, and testing previously untested updates as singletons adds no attacker rejections on those replays. On re-executed canonical runs the condition holds less often; there, copied attackers are routed away from the test by their extreme density, and under strong heterogeneity the test sometimes removes Stage-1-accepted updates. Random subsets of the same size show that Multi-Krum’s adversary-free loss at $\alpha = 0.01$ reflects which updates it selects rather than how many. These findings support reporting benign cost, attack detection, and attack success together, with implementation boundaries and conditional arguments stated explicitly.

ARTIFACT AVAILABILITY

The accompanying artifact contains cell and per-unit results, failure accounting, checkpoint summaries, and scripts that regenerate the manuscript tables and figures. The project repository is <https://github.com/Bygokcen/fed-mdbscan-g>. Raw training artifacts and frozen sources remain in the local research archive and are not included in a repository clone. The supplied summaries permit checking the reported arithmetic; reproducing the historical identity and checkpoint audits also requires that archive.

ACKNOWLEDGMENT

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the authors used generative AI tools for language editing and manuscript formatting assistance, and for developing and checking analysis code. All experimental results derive from the archived run records and analysis scripts. The authors reviewed and edited all AI-generated content, taking full responsibility for the content of the publication.

AUTHOR CONTRIBUTIONS

S. G. Özden: methodology, software, validation, formal analysis, investigation, data curation, visualization, writing—original draft. K. Sarıkaya: conceptualization, supervision, writing—review and editing.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, ser. PMLR, vol. 54, 2017, pp. 1273–1282.
- [2] M. Fang, X. Cao, J. Jia, and N. Z. Gong, "Local model poisoning attacks to Byzantine-robust federated learning," in *Proc. 29th USENIX Security Symp.*, 2020, pp. 1623–1640.
- [3] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *Proc. 23rd Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, ser. PMLR, vol. 108, 2020, pp. 2938–2948.
- [4] C. Xie, K. Huang, P.-Y. Chen, and B. Li, "DBA: Distributed backdoor attacks against federated learning," in *Proc. 8th Int. Conf. Learning Representations (ICLR)*, 2020.
- [5] P. Blanchard, E. M. E. Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 119–129.
- [6] D. Yin, Y. Chen, K. Ramchandran, and P. L. Bartlett, "Byzantine-robust distributed learning: Towards optimal statistical rates," in *Proc. 35th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 80, 2018, pp. 5650–5659.
- [7] X. Cao, M. Fang, J. Liu, and N. Z. Gong, "FLTrust: Byzantine-robust federated learning via trust bootstrapping," in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2021.
- [8] T. D. Nguyen, P. Rieger, H. Chen, H. Yalame, H. Möllering, H. Fereidooni, S. Marchal, M. Miettinen, A. Mirhoseini, S. Zeitouni, F. Koushanfar, A.-R. Sadeghi, and T. Schneider, "FLAME: Taming backdoors in federated learning," in *Proc. 31st USENIX Security Symp.*, 2022, pp. 1415–1432.
- [9] G. Baruch, M. Baruch, and Y. Goldberg, "A little is enough: Circumventing defenses for distributed learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 8632–8642.
- [10] V. Shejwalkar and A. Houmansadr, "Manipulating the Byzantine: Optimizing model poisoning attacks and defenses for federated learning," in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2021.
- [11] A. K. Singh, A. Blanco-Justicia, and J. Domingo-Ferrer, "Fair detection of poisoning attacks in federated learning on non-i.i.d. data," *Data Mining and Knowledge Discovery*, vol. 37, no. 5, pp. 1998–2023, 2023.
- [12] P. Ye, Y. Li, K. He, Q. Li, T. Qin, X. Wang, K. Yang, C. Zhang, and J. Chen, "Breaking the illusion: A critical study of backdoor defense in federated learning with non-IID data," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 13 264–13 279, 2025.
- [13] A. A. Bellachia, M. A. Bouchiha, and Y. Ghamri-Doudane, "SoK: Understanding backdoor attacks and defenses in federated learning," *Expert Systems with Applications*, vol. 325, p. 132558, 2026.
- [14] O. Touat and S. Bouchenak, "Towards robust and bias-free federated learning," in *Proc. 3rd Workshop on Machine Learning and Systems (EuroMLSys)*, 2023.
- [15] O. Ajibuwah and O. T. Odeyomi, "On the hidden costs of fairness in Byzantine-robust decentralized federated learning under data heterogeneity," in *Proc. IEEE Int. Performance, Computing, and Communications Conf. (IPCCC)*, 2025.
- [16] A. Djemaa, D. Djenouri, and P. Legg, "Heterogeneity-aware poisoning attacks and mitigation in federated learning: A comprehensive survey and taxonomy," *Electronics*, vol. 15, no. 13, p. 2876, 2026.
- [17] K. Pillutla, S. M. Kakade, and Z. Harchaoui, "Robust aggregation for federated learning," *IEEE Trans. Signal Process.*, vol. 70, pp. 1142–1154, 2022.
- [18] S. P. Karimireddy, L. He, and M. Jaggi, "Byzantine-robust learning on heterogeneous datasets via bucketing," in *Proc. 10th Int. Conf. Learning Representations (ICLR)*, 2022.
- [19] Y. Allouah, S. Farhadkhani, R. Guerraoui, N. Gupta, R. Pinot, and J. Stephan, "Fixing by mixing: A recipe for optimal Byzantine ML under heterogeneity," in *Proc. 26th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, ser. PMLR, vol. 206, 2023, pp. 1232–1300.
- [20] P. Rieger, T. D. Nguyen, M. Miettinen, and A.-R. Sadeghi, "DeepSight: Mitigating backdoor attacks in federated learning through deep model inspection," in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2022.
- [21] E. Kabir, Z. Song, M. R. U. Rashid, and S. Mehnaz, "FLShield: A validation based federated learning framework to defend against poisoning attacks," in *Proc. IEEE Symposium on Security and Privacy (SP)*, 2024, pp. 2572–2590.
- [22] P. Rieger, T. Krauß, M. Miettinen, A. Dmitrienko, and A.-R. Sadeghi, "CrowdGuard: Federated backdoor detection in federated learning," in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2024.
- [23] T. Krauß and A. Dmitrienko, "MESAS: Poisoning defense for federated learning resilient against adaptive attackers," in *Proc. 2023 ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2023, pp. 1526–1540.
- [24] Z. Chen, F. Wang, Z. Zheng, X. Liu, and Y. Lin, "Fedward: Flexible federated backdoor defense framework with non-IID data," in *Proc. IEEE Int. Conf. Multimedia and Expo (ICME)*, Brisbane, Australia, 2023, pp. 348–353.
- [25] J. Yuan, Q. Zhang, N. Chen, S. Chen, and B. Xu, "A multi-granularity clustering approach for federated backdoor defense with the Adam optimizer," in *Proc. 34th Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2025, pp. 6931–6939.
- [26] M. V. Ramana, C. Nagabharani, S. G. Krishna, V. V. Mojesh, and A. Malathi, "Adaptive local density-based defense against poisoning attacks in federated learning," in *Proc. 5th Int. Conf. Computer Networks, Big Data and IoT (ICCB)*, Dubai, United Arab Emirates, 2026.
- [27] J. Qian, Y. Zhou, X. Han, and Y. Wang, "MDBSCAN: A multi-density DBSCAN based on relative density," *Neurocomputing*, vol. 576, p. 127329, 2024.
- [28] P. Viola and M. J. Jones, "Robust real-time face detection," *International Journal of Computer Vision*, vol. 57, no. 2, pp. 137–154, 2004.
- [29] M. Kloft and P. Laskov, "Security analysis of online centroid anomaly detection," *Journal of Machine Learning Research*, vol. 13, pp. 3681–3724, 2012.
- [30] Q. Li, D. Wu, D. Zhou, C. Lin, S. Liu, C. Wang, and C. Shen, "Robust adversarial defenses in federated learning: Exploring the impact of data heterogeneity," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 6005–6018, 2025.
- [31] V. Shejwalkar, A. Houmansadr, P. Kairouz, and D. Ramage, "Back to the drawing board: A critical evaluation of poisoning attacks on production federated learning," in *Proc. IEEE Symp. Security and Privacy (S&P)*, 2022, pp. 1354–1371.
- [32] G. Abad, M. Krček, S. Koffas, B. Tajalli, M. Arazzi, R. Riaño, X. Xu, Z. Liu, A. Nocera, and S. Picok, "SoK: The last line of defense: On backdoor defense evaluation," *arXiv preprint arXiv:2511.13143*, 2025.
- [33] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [34] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms," *arXiv preprint arXiv:1708.07747*, 2017.
- [35] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A public domain dataset for human activity recognition using smartphones," in *Proc. 21st European Symp. Artificial Neural Networks (ESANN)*, 2013, pp. 437–442.
- [36] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009. [Online]. Available: <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>
- [37] T.-M. H. Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," *arXiv preprint arXiv:1909.06335*, 2019.